Introduction

At present, aircraft departing an airport are successively cleared to some intermediate altitudes before they reach their desired cruise flight level. These so-called climb steps are mainly enforced by the air traffic control (ATC) in order to maintain separation with conflicting traffic flows. A similar situation occurs for arriving traffic, which descend progressively through intermediate altitudes from their cruise flight level to the altitude at which the Instrument Landing System (ILS) is intercepted.

Continuous Climb Operations (CCO) aim to enable trajectories that, from take-off, follow a continuous and optimal climb to the cruise altitude without requiring intermediate low-altitude steps. This approach offers several benefits, including reduced fuel consumption and noise. However, it also raises several challenges regarding its implementation, particularly at busy airports where interactions between arriving and departure aircraft are significant.

Similarly, Continuous Descent Approaches (CDA) or more generally **Continuous Descent Operations (CDO)**, offer similar benefits for arriving traffic. They have been the subject of research over the past decade and have been applied successfully at some airports [1-3]. However, a major drawback of CCOs and CDOs is their potential impact on terminal airspace capacity, at ATC may need to apply larger separation margins due to greater uncertainty in trajectory prediction. Consequently, at many airports, CCOs and CDOs are only applied in off-peak hours, such as at night.

In this project, commissioned by the Air Navigation Service Provider (ANSP), you will **assess the impact of implementing Continuous Descent Operations (CDO) for arrivals at Barcelona–El Prat Airport (LEBL)**. The study is intended to provide the ANSP with an initial assessment of the operational implications of introducing CDO procedures for arrivals at the airport. The results of this introductory study will support the ANSP in evaluating the feasibility of subsequent flight trials and the potential implementation of CDO in its arrival procedures.

Special requirements

- Before the Kick-off Meeting (KOM), please read this Statement of Work carefully.
- For this Work Package (WP), you will be required to develop a program in **MATLAB or Python**. Therefore, you will need access to computers with the selected programming language and the necessary development environment already installed.

Statement of Work

For this project, you must work in **teams of four students** (ITA group) and shall complete the following activities:

1. **Conduct a literature review on CDOs and arrival sequencing techniques. Attention should be given to their advantages, drawbacks, implementation requirements, and current deployment status.**

For this first activity, you shall identify and review relevant bibliographic references on CDOs, aircraft trajectory optimisation, and Arrival Managers (AMAN). The use of **high-quality sources**, such as books, journal articles, and conference proceedings, is strongly encouraged. As part of this task, you

shall explain the background and operational concept of CDOs, their main benefits and drawbacks, the technology required to support their implementation, and their current implementation status.

2. Implement a CDO trajectory simulator

Aircraft of different types have different performance characteristics. To analyse these differences, your group will simulate the trajectories of descending aircraft using a MATLAB/Python program. The following five aircraft types shall be considered: **B767-300ER, B777-300, B737, A320-212, and A319-131**.

For the trajectory simulations, assume that all aircraft fly at **idle thrust while minimising their rate of descent (ROD)** until they intercept the Initial Approach Fix (IAF). All aircraft shall intercept the IAF at **6,000 ft**.

The following additional assumptions shall be considered:

- The atmosphere shall be modelled according to the **International Standard Atmosphere (ISA)**.
- At the IAF, the aircraft weight shall **not exceed the Maximum Landing Weight (MLW)**.

For further information on the performance models to be used for the trajectory simulations, see **Annex A**. The parameters for the aircraft considered in this project are provided in **Annex B**.

3. Analyse a multiple-CDO arrival scenario at LEBL (RWY 24L)

Consider that the following aircraft are planned to arrive at **RWY 24L at LEBL**:

Aircraft type	STAR	Weight at IAF
B767-300ER	ALBER1Z	80% MLW
B737	PUMAL1Z	100% MLW
B777-300	MARTA3Z	100% MLW
B767-300ER	MATEX3Z	80% MLW
A319-131	LOBAR2W	80% MLW
A320-212	CASPE2W	100% MLW

Assume that the altitude of each aircraft at the STAR entry point is the altitude required for the aircraft to **intercept the IAF at 6,000 ft** under the CDO assumptions defined above.

Two scenarios shall be analysed:

- **SCENARIO 1: Simultaneous arrivals at the STAR entry points**
    - Calculate the arrival time of each aircraft at the IAF, assuming that all aircraft arrive at the first waypoint (WP) of the STAR at **11:45:00 AM**.
    - Considering that both **runway throughput and safety levels** must be maintained, assess the suitability of this scenario. If necessary, propose potential **tactical and/or pre-tactical measures** to achieve a more efficient arrival sequence.

- **SCENARIO 2: Optimisation of arrival sequencing at the IAF**
    - To optimise runway throughput while maintaining safety levels, the objective of the approach air traffic controllers (ATCOs) is to establish a **constant time separation between consecutive aircraft arriving at the IAF**. To achieve this objective, assume that approach controllers can coordinate with en-route controllers to establish the arrival time of each aircraft at the STAR entry point.
    - Propose **new arrival times at the STAR entry point** based on the following considerations:
        - The **time separation between consecutive aircraft at the IAF shall be exactly 2 minutes**.
        - The arrival time of each aircraft at the STAR entry point may be advanced or delayed with respect to the time defined in Scenario 1. **No time adjustment shall be applied to the first aircraft arriving at the IAF.**
        - Among the multiple possible combinations, select the one that **maintains the same order of arrival at the IAF**.
    - Discuss the advantages and disadvantages of this scenario.

For each scenario, the **altitude of each aircraft at the STAR entry point** shall be included in the results of the analysis.

4. **Assess the suitability of the simulated trajectories with respect to the current STAR altitude constraints**

All altitude constraints shall be obtained from the **STAR chart published by ENAIRE in the AIP**. Compare the simulated trajectories with the published altitude constraints and assess their compliance. Any deviations or potential operational issues shall be identified and discussed.

5. **Write a paper presenting the findings and recommendations**

The paper shall present the main findings of the project and provide recommendations based on the analyses performed. It shall include an **abstract, introduction, description of the implemented algorithm, presentation of the arrival sequencing analysis, discussion of compliance with the STAR altitude constraints, conclusions, and references**. All references shall be properly cited in the text.

All decisions, assumptions, methods, and conclusions presented in the paper and throughout the project shall be properly justified.

The paper shall follow the template provided on Atenea. The maximum length is **8 pages, including references**.

IMPORTANT: Papers that do not follow the required template structure and/or are written in a language other than **English** will be automatically rejected.

WP evaluation

The final score of this WP will be mainly based on the quality of the final report. Individual progress in class may be taken into consideration under special circumstances.

Annex A: Aircraft performance models

In order to model aircraft performance, the Eurocontrol's base of aircraft data (BADA) shall be used [4]. BADA is a total energy aircraft performance model developed by Eurocontrol, which characterises the performances of the aircraft by a set of parameters stored in a database.

According to BADA, the drag coefficient is defined as:

$$CD=CD_0+CD_2CL^2$$

where CD_0 and CD_2 are two parameters specified in BADA as a function of the aircraft and its flaps/slats configuration (approach or clean) and CL is the lift coefficient.

The descent thrust (idle thrust) is computed as a function of the maximum thrust and the pressure altitude (h_p) expressed in ft.

If h_p>$h_{p,desc}$:

$$T_{desc}=CT_{Desc,high}\cdot T_{max}$$

Else:

Clean configuration: $T_{desc}=CT_{Desc,low}\cdot T_{max}$

Approach configuration: $T_{desc}=CT_{Desc,app}\cdot T_{max}$

where $CT_{Desc,high}$, $CT_{Desc,low}$ and $CT_{Desc,app}$ are thrust performance parameters specified in BADA and the maximum thrust is computed as:

$$T_{max}=C_{T1}\left(1-\frac{h_p}{C_{T2}}+C_{T3}{h_p}^2\right)$$

where C_{T1}, C_{T2} and C_{T3} are also aircraft dependent performance parameters specified in BADA.

Consider that all aircraft enter in approach configuration at h_p=$6000ft$. Note that when the aircraft is in the approach configuration the values of CD0 and CD2 also change due to the modification of the aerodynamic surfaces of the wing.

The equations of the balance of forces must assume that the angle of attack is small, the thrust vector is aligned with the aircraft longitudinal symmetry axis and the wind situation is calm.

The thrust specific fuel consumption (η), is modeled in [4] as a function of the true airspeed:

$$\eta=C_{F1}\left(1+\frac{v}{C_{F2}}\right)$$

where $C_{F1}$and C_{F2} are aircraft dependent parameters. Finally, the fuel flow (FF) is computed as:

$$FF=\eta T$$

Annex B: Aircraft performance data

In the following table, all the BADA coefficients of the different aircraft types considered in the study and extracted from [4] are given:

	B767-300ER	B777-300	B737	A320-212	A319-131
Max Landing Weight (tons)	0.145150E+03	0.237680E+03	0.51710E+02	0.64500E+02	0.61000E2
Max Weight (tons)	0.20410E+03	0.29930E+03	0.70800E+02	0.77000E+02	0.70000E+02
Max. Payload (tons)	0.46500E+02	0.64900E+02	0.16920E+02	0.21500E+02	0.17000E+02
S (m^2)	0.28350E+03	0.42804E+03	0.12465E+03	0.12260E+03	0.12260E+03
$CD_{0,\,app}$	0.14000E-01	0.17300E-01	0.27000E-01	0.24200E-01	0.28400E-01
$CD_{2,\,app}$	0.49000E-01	0.48400E-01	0.44100E-01	0.46900E-01	0.37600E-01
$CD_{0.\,clean}$	0.17400E-01	0.15700E-01	0.23500E-01	0.24000E-01	0.28000E-01
$CD_{2,\,clean}$	0.45900E-01	0.42000E-01	0.44500E-01	0.37500E-01	0.31000E-01
$h_{p,desc}$ (ft)	26418	36122	30152	12398	27726
$CT_{Desc,high}$	0.64359E-1	0.44239E-1	0.36336E-1	0.45711E-1	0.83084E-1
$CT_{Desc,low}$	0.55988E-1	0.41065E-1	0.53395E-1	0.27207E-1	0.51765E-1
$CT_{Desc,app}$	0.12475	0.92921E-1	0.16440	0.13981	0.14767
C_{T1} (N)	.35167E+06	.42577E+06	.14573E+06	.13605E+06	.13900E+06
C_{T2} (ft)	.44673E+05	.48987E+05	.55638E+05	.52238E+05	.58900E+05
C_{T3} ($1/ft^2$)	.10129E-09	.66146E-10	.14200E-10	.26637E-10	.57200E-14
C_{F1} (kg/(min·kN))	.54005E+00	.87843E+00	.94680E+00	.94000E+00	.68800E+00
C_{F2} (kt)	.55782E+03	.36897E+04	.10000E+15	.10000E+06	.16700E+04

References

[1] Clarke, John-Paul, Ho, Nhut, Ren, Liling, Brown, John, Elmer, Kevin, Tong, Kwok-On and Watt, Joshep, *Continuous Descent Approach: Design and Flight Test for Louisville International Airport*, Journal of Aircraft, Vol. 41, No. 5, pp. 1054-1066, September-October 2004.

[2] Jin, Li, Cao, Yi and Sun, Dengfeng, *Investigation of Potential Fuel Savings Due to Continous-Descent Approach*, Journal of Aircraft, In Press, February 2013

[3] Reynolds, Tom, Ren, Liling, Clarke, John-Paul, Bruke, Andrew and Green, Mark, *History, Development and Analysis of Noise Abatement Arrival Procedures for UK Airports*, AIAAA 2005-7395, AIAA 5th Aviation Technology, Integration and Operations Forum (ATIO), Arlington, VA, 26-28 September 2005

[4] Eurocontrol Experimental Centre, *User Manual for the Base of Aircraft Data (BADA) Revision 3.10*, EEC Technical/Scientific Report No. 12/04/10-45, April 2012.